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Physical Performance Tests of a Tandem Ultra-High-Speed High-Gas-Volume-Fraction Pump

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Abstract

Conventional multiphase pump technologies typically handle intake gas volume fractions (GVFs) of 75%. Recent work has shown development of an Ultra-High-Speed High GVF Pump (UHS HGVFP) with significant boost pressure at 90% intake GVF. The current work discusses the physical performance test results of a tandem UHS HGVFP and showcases pump behavior with increased number of modules. Knowledge from such work is beneficial in multiphase pump sizing for production in high gas-oil ratio wells.

A 4 in. housing outer diameter, tandem (double module) UHS HGVFP was tested at a rotational speed of 10,000 rev/min. Air and water were the working fluids, and liquid volume flow rates varied from about 550 to 1980 barrels per day (B/D). The gas flow rates were adjusted accordingly to achieve a 0% to 90% intake GVF range. For each test data, pressures and temperatures at pump intake and pump discharge and gas lines were measured, with corresponding pump boost pressures computed. Pump performance was plotted and results compared with those of a single-module UHS HGVFP at same GVFs.

The results showed a steady decrease in boost pressure as the intake GVF varied from 0% (liquid-only tests) to 90%. For the same boost pressures, the magnitudes of liquid volume flow rate delivered by the pump decreased with increasing intake GVF and vice versa. The tandem UHS HGVFP test results were compared to those of a previously tested single-module UHS HGVFP when pumping only water. For the same liquid volume flow rates and intake pressures, the boost pressure from the tandem-module was, on average, about two (2) times those of the single-module UHS HGVFP. This indicates that for liquid only tests, doubling the number of stages also doubles the pump boost pressure. For intake GVFs from 15% to 90%, the tandem UHS HGVFP produced about 1.24 to 1.88 times the boost pressure of a single-module UHS HGVFP. Lower boost pressure ratios were observed as intake GVF approached 90%. These results indicate that at higher intake GVFs, doubling the number of modules does not double the boost pressure boost capability of the pump. Knowledge gained from these tests are vital for successful multiphase performance sizing of the UHS HGVFP ahead of field installation and trial.

At low intake GVFs, a tandem (2-module) UHS HGVFP can produce double the pressure boost of a single-module UHS HGVFP, but is not the case at higher intake GVFs up to 90%. Knowledge of such

performance trends are crucial for proper sizing of the UHS HGVFP. Results from this study indicate that a series of multiple-module UHS HGVFP has superior capabilities to handle multiphase flows in wells with high gas-oil-ratios.

Introduction

Electric submersible pumps (ESP) are widely used artificial lift systems in the oil and gas industry. The major sections of an ESP pumping system are assembled as stages, which are each made up of a rotary impeller and a stationary diffuser. The fluid moving through the ESP is given kinetic energy by the rotary action of the impeller. The diffuser on the other hand transforms the fluid kinetic energy exiting the impeller into pressure energy as the fluid is subsequently discharged into the inlet of the impeller of the next stage. Typical operating rotational speeds of conventional ESPs is 4,200 rev/min. In more recent years, ESP operating speeds have increased to a range above 4200 to but less than 10,000 rev/min (Amsidom et al., 2018; Lastra et al., 2019). These systems, which have smaller housing diameters, or called slim systems, have handled mostly liquid-only fluids in their field applications. In pushing ESP liquid-handling operating envelope, ultra-high-speed ESPs have been developed (Zamanuri et al., 2019), which operate at rotational speeds of 10,000 rev/min and beyond. The much higher rotational speeds significantly increases the liquid boost pressure as well as liquid volume flow rate. These ultra-high-speed pumps have paved the way for installing ESPs with even smaller housing diameters to effectively produce reservoir fluids from slim wells.

ESPs typically handle liquid-only flows satisfactorily without degradation in performance. When handling gas-liquid flows, ESP performance decreases substantially when the free gas content entering the pump surpasses a certain magnitude of gas volume fraction (GVF). The GVF is the ratio of free gas volume to the total volume of free gas and liquid at a given temperature and pressure. The gas handling capability of ESPs depends on the pump-type structure (radial, mixed-flow or axial). Extreme performance deterioration of ESP pressure boosting capability due to high GVF gives rise to gas-lock (Hua et al., 2012). These conditions are associated with situations in field pumping operations, where the ESP is operational, but no reservoir fluid production is received at the surface. These challenges can be mitigated in conventional artificial lift production by installing gas separators, gas handlers or multiphase pumps upstream of the ESP, which operate at similar conventional rotational speeds. Intake GVF limitations of conventional multiphase gas handlers is 75% (Bagci et al., 2010). This has resulted in the need to develop pumps that can handle much higher intake GVFs for use in high wells with high Gas-Oil-Ratio (GOR).

Ejim et al. (2025) demonstrated the physical capability of an Ultra High-Speed High Gas Volume Fraction Pump (UHS HGVFP) that can handle GVF up to 90%. In their work, a single module rotor-stator pair was tested to determine the performance characteristics of the pump. The current study is an extension of that work, since the pump modules will be assembled in series in the field application to provide the required fluid boost pressure when pumping multiphase mixtures with very high gas-content. The current work focuses on obtaining the physical performance behavior of a tandem (two pump modules in series) UHS HGVFP. The aim is to establish a deeper understanding of the boost pressure capabilities of a series configuration of the UHS HGVFP. The overall objective is to significantly improve pump application design and sizing in preparation for field trials when pumping high GVF multiphase flows at a given pump setting depth in a well.

Tandem UHS HGVFP Architecture

To have the UHS HGVFP in tandem, each single module of the pump was connected to the other by means of a suitable shaft coupling. Each module of the UHS HGVFP had a housing outer diameter of 4 in. The modules are made up of a rotor that has threads with externally shaped helical profile. These rotors are enclosed by a stator with helically shaped internal thread profile. The gap between the maximum rotor tip outer diameter and the minimum stator inner diameter is sufficiently large to produce flows with sand

concentrations up to 3.51 lbm/barrel. Each rotor-stator pair has a dedicated thrust module to handle the downthrust loads developed by the UHS HGVFP during pumping operation. The rotor and stator profiles were designed to extensively increase fluid turbulence, and vortices. This significantly enhances mixing of the gas and liquid streams leading to a substantial reduction in the susceptibility of gas lock. The drive mechanism for the UHS HGVFP is a permanent magnet motor with high efficiency and incorporated with a heat exchanger mechanism for active cooling of the motor during operation (Shakirov et al., 2019). The pump is designed for rotational speeds up to 12000 rev/min. To ensure the pump meets the required vibration limits, the rotors are properly balanced. Furthermore, to prevent occurrence of resonance during pump operation, the UHS HGVFP is assembled in a modular configuration. Finally, the combination of all the features of the pump results in a much shorter pump module compared to conventional systems of the same boost pressure capacity. This aids in equipment handling and transportation with the potential of faster field installation and retrieval from a well.

Test Layout and Procedure

The setup for testing the tandem UHS HGVFP is shown in Figure 1. The test fluids were air and water. Sufficient liquid was held in the water tank to ensure stable liquid supply throughout the test. The water tank was also used as the discharge enclosure for the air-water mixture exiting the pump. The liquid from the water tank flows into a heat exchanger system (not shown) to maintain the liquid temperature. The liquid is then drawn by the circulation pump and flows through a flow meter and regulating valve on the liquid line to a mixing device. The air supply line is also connected to the mixing device. This supply line includes a compressor, gas mass flow meter, temperature/pressure sensors, and a gas flow regulating valve to maintain control of the gas flow. The intake of the UHS HGVFP is fitted with pressure and temperature sensors and connected to the mixing device. The drive motor for the tandem UHS HGVFP is directly coupled upstream of the pump. The motor is connected to a Variable Speed Drive (VSD) (not shown) for easy control and change of the rotational speed of the pump-motor system. The pump discharge line is fitted with pressure and temperature sensors to measure pump discharge parameters. The fluid mixture discharged from the pump flows through a discharge line into the water tank. The list of measurement equipment is shown in Table 1.

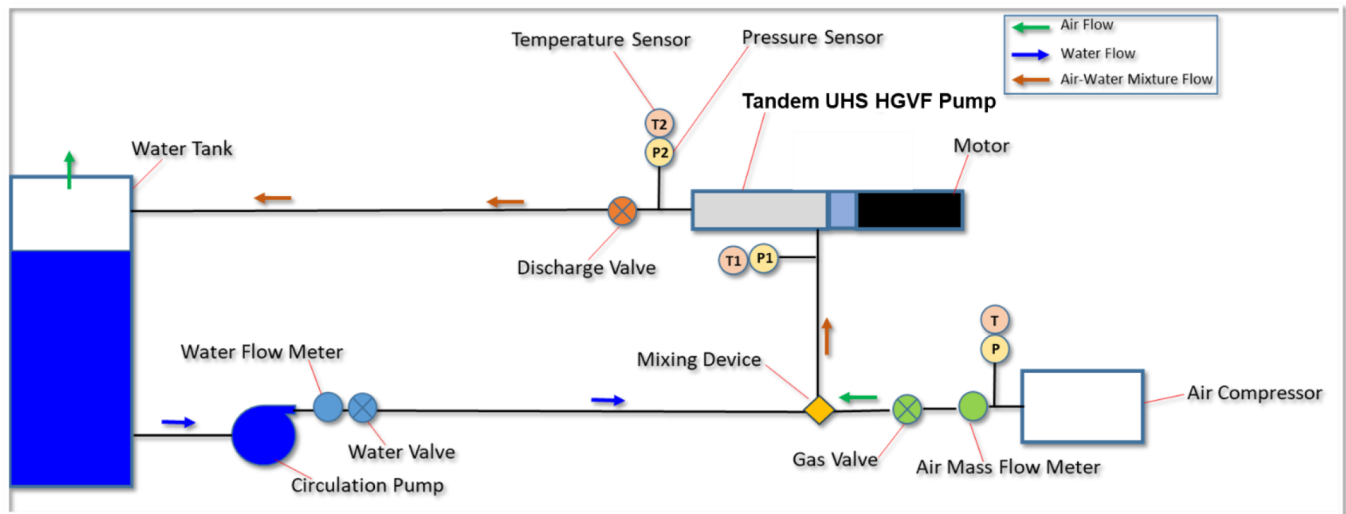


Figure 1—Schematic setup of the tandem UHS HGVFP test.

The air supply line is also connected to the mixing device. This supply line includes a compressor, gas mass flow meter, temperature/pressure sensors, and a gas flow regulating valve to maintain control of the gas flow. The intake of the UHS HGVFP is fitted with pressure and temperature sensors and connected to the mixing device. The drive motor for the tandem UHS HGVFP is directly coupled upstream of the pump. The motor is connected to a Variable Speed Drive (VSD) (not shown) for easy control and change of the rotational speed of the pump-motor system. The pump discharge line is fitted with pressure and temperature sensors to measure pump discharge parameters. The fluid mixture discharged from the pump flows through a discharge line into the water tank. The list of measurement equipment is shown in Table 1.

Table 1—Measurement equipment list (FS = Full Scale)

Equipment	Range	Accuracy
Liquid Flowmeter	0 – 8200 B/D	0.9% relative error limit
Thermal Mass Flowmeter	0 – 9100 B/D	± 3.00% FS
Intake Pressure Gauge	0 – 130 psi	± 0.25% FS
Discharge Pressure Gauge	0 – 870 psi	± 0.25% FS
Resistance Temperature Detector (RTD)	0 - 250°F	± 0.50% FS

The test commenced by first closing the valve on the air line and fully opening the valve on the water line. The rotational speed of the tandem UHS HGVFP was gradually increased to 10,000 rev/min by means of the VSD. Using the water valve, the water flow rates were adjusted to desired magnitudes prior to flow stabilization and subsequent measurements. The flow rates for tests using only water varied from 1276 to 1980 B/D. When the gas-liquid tests were to be performed, air was injected into the system gradually to increase the intake GVF. For each data point, the tests were performed at relatively constant intake pressure, with varying water and air intake volume flow rates resulting in a GVF range from 0% to 90%. Fifty-nine (59) data points in total were collected for the combined water only and air-water tests, with a liquid volume flow rate range of 550 to 1980 B/D. This facilitated developing a performance correlation for the pump boost pressure. The average fluid pressure and temperature at the intake of the tandem UHS HGVFP throughout the test was about 35 psig and 74°F, respectively. After data measurement, the pressure boost variation of the tandem pump with liquid volume flow rate was plotted. Subsequently, the variation of the pump pressure boost with liquid flow rate for different intake GVFs was also plotted. The data for the tandem system was compared to corresponding data for a previously tested single module system to ascertain pressure boost differences.

Pumping Power Equation for Gas-Liquid Mixtures

The power transferred by an energy-adding device, such as pumps to a fluid, may be determined by applying the steady flow energy equation from fundamental thermodynamics (Cengel et al., 2019). The thermodynamics is essential since for the very high intake GVFs, applicable to the current study, gas compressibility effects results in a non-isothermal pumping process even though the gas specific heat capacity is much smaller than that of the liquid. Neglecting kinetic energy and potential energy differences, which are very very small compared to the pressure terms, it can be shown fundamentally that the useful power, P_{mix} , given to a fluid mixture can be expressed as shown in Equation (1):

$$P_{mix} = \left\{ (P_2 - P_1) Q_L \right\} + \left\{ P_1 Q_{G1} \left[\frac{n}{n-1} \left[\left(\frac{P_2}{P_1} \right)^{\left(\frac{n-1}{n} \right)} - 1 \right] \right] \right\} \quad (1)$$

Where,

Pump intake and discharge conditions are denoted by subscripts 1 and 2, respectively

P_1 = Absolute pressure of fluid mixture at pump intake

P_2 = Absolute pressure of fluid mixture at pump discharge

Q_L = Volume flow rate of the liquid

Q_{G1} = Volume flow rate of the gas at pump intake conditions

n = Polytropic index (for polytropic processes $n > 1$)

In Equation (1), the first term in the curly ({}) brackets is the hydraulic power given to the liquid to raise its energy level (or pressure). The second term is the power given to the gas stream for polytropic compression conditions.

Results

The test results presentation is divided into three segments. The first segment discusses the boost pressure variation of the tandem module UHS HGVFP. The second segment shows the variation of the tandem module when the intake GVF varies from 0% to 90%. The third segment compares the performance behavior of the tandem module with those a previously tested single module UHS HGVFP when handling multiphase flow with intake GVF from 0% to 90%. The data has been summarized significantly for the write-up to show the main performance behavior of the pumps.

Tandem Module Liquid-Only Tests

Figure 2 shows the variation of boost pressure with liquid volume flow rate for the tandem module UHS HGVFP when pumping only liquid (GVF = 0%). The pump develops a boost pressure of about 780 psi at approximately 1300 B/D, and about 540 psi at 1980 B/D. This decreasing trend in boost pressure with increasing flow rate (and vice versa) is also a characteristic trend in conventional ESPs. The current trend is also similar to observations by Ejim et al., (2025). From the figure, boost pressure decreases by about 0.35 psi for each barrel increase in liquid flow rate.

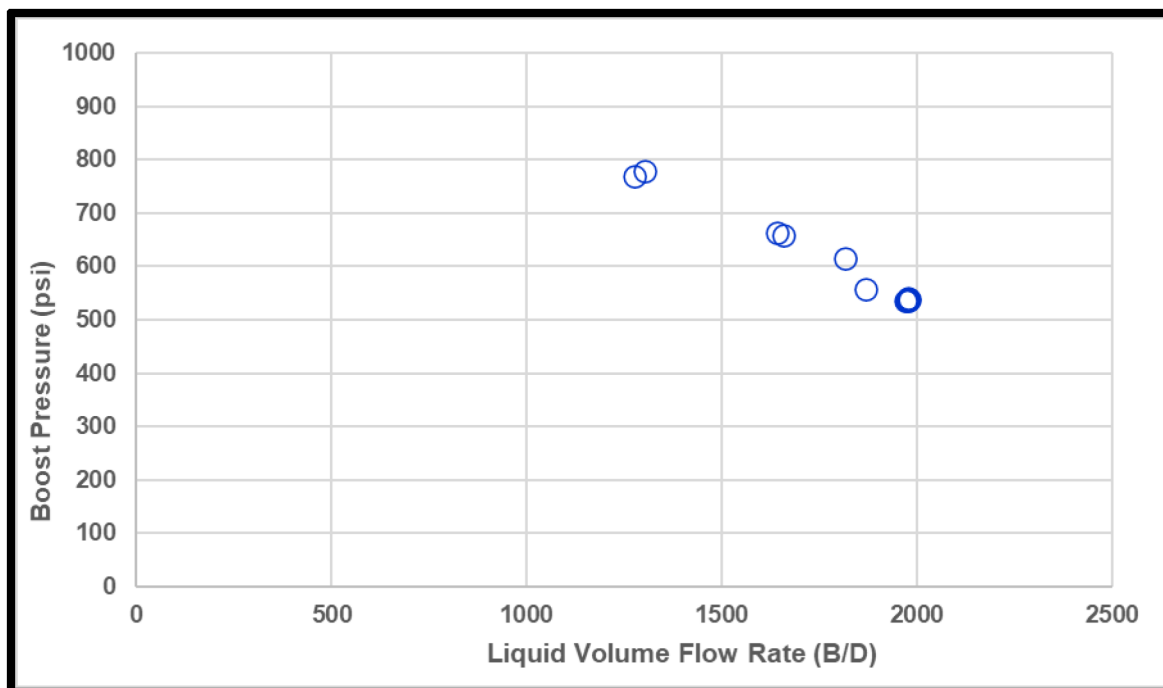


Figure 2—Boost pressure vs liquid flow rate for tandem module at 10,000 rev/min (GVF = 0%; intake pressure = 35psi).

Tandem Module Multiphase Tests

The variation of pressure boost with liquid volume flow rate for intake GVFs of 15%, 30%, 45%, 60%, 75% and 90% is shown in Figure 3 with the liquid-only tests (0% GVF) included for comparison. For each intake GVFs greater than zero, the boost pressure provided by the tandem pump also decreases with increasing liquid volume flow rate. The boost pressure variation with flow rate for each plot is close to a negatively sloped linear curve similar to the trend in Figure 2 for the liquid-only tests.

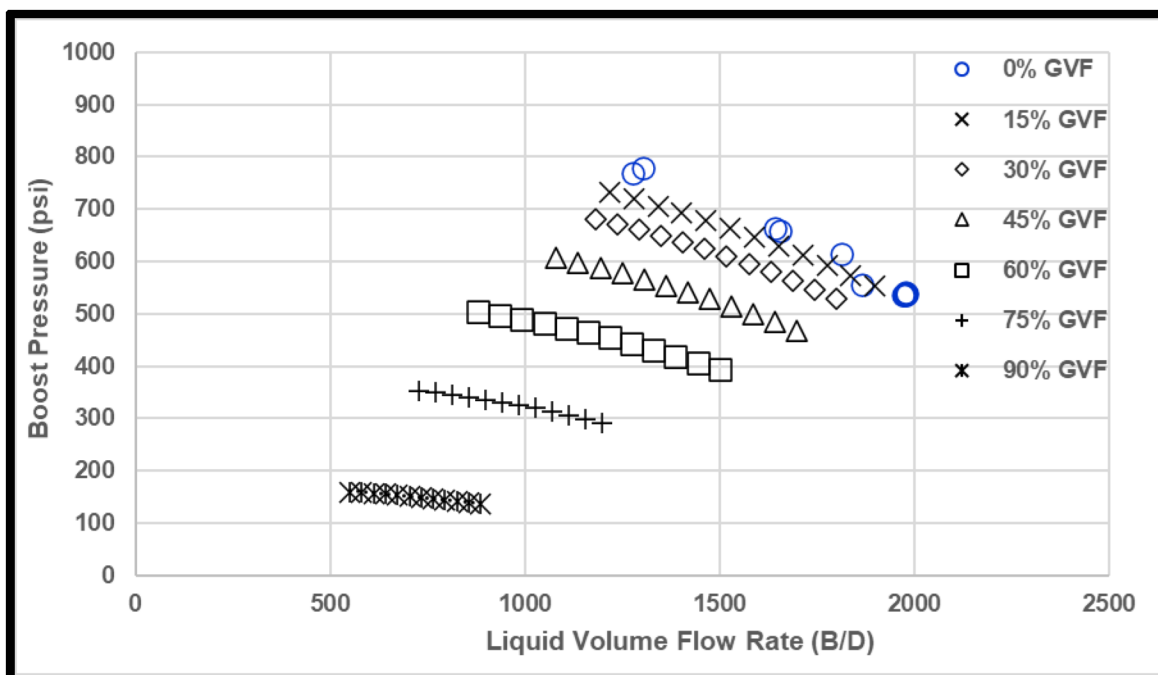


Figure 3—Boost pressure vs liquid flow rate for the tandem module at different intake GVFs and 10,000 rev/min (intake pressure = 35 psi).

An observation from Figure 3 is also the decrease in slope of the pressure boost with increasing intake GVF. As the intake GVF increases towards 90% the boost pressure still varies inversely with liquid volume flow rate, but the change in boost pressure becomes very, very small. Hence, at high intake GVFs, for a significant pump boost pressure to meet a target pressure and liquid volume flow rate, additional modules will be sized accordingly to meet the operating requirement.

Another trend in Figure 3 is the decrease in liquid flow rate capability the pump can deliver as the intake GVF increases. Extrapolating the liquid only (0% GVF), it is possible to attain flow rates just over 3500 B/D. However, at 90% intake GVF, attaining flow rates of 2800 B/D is not feasible. These effects are typical of pumps when handling gas-liquid mixtures, which is mainly due to the expansion of the gas phase, which occupies the flow channels within the pump. Whereby these effects are more pronounced in conventional tandem ESPs, the tandem module UHS HGVFP shows resilience to gas effects by maintaining a significant positive boost pressure even at 90% intake GVF, indicating no gas-lock phenomenon occurs.

Continuing from the discussion above, also observed is the decrease in boost pressure with increasing intake GVF for a given liquid volume flow rate. For instance, as mentioned previously, at 0% GVF, the pump is able to deliver 1300 B/D of liquid at a boost pressure of 780 psi. However, as the intake GVF increases to 15%, 30%, 45% and 60%, the tandem module UHS HGVFP is only able to provide boost pressures of 726 psi, 670 psi, 580 psi, and 450 psi, respectively at a liquid volume flow rate of 1300 B/D. At 75% and 90% intake GVF, it can be estimated that the boost pressure is about 280 psi and 100 psi, respectively at the same liquid volume flow rate.

Tandem and Single Module Performance Comparison at 0% Intake GVF

The boost pressure variation with liquid volume flow rate for the tandem module and a previously tested single module are shown in Figure in 4. Comparing the boost pressures of both systems at the same liquid volume flow rate, the tandem module develops on average about two times the boost pressure of the single module. Data for one-half the boost pressure of the tandem system is also plotted in Figure 4. It is observed that the data points overlap on the data for the single module UHS HGVFP. From fundamental literature of pumps handling liquids only, the boost pressure of pumps in series is additive. Given that the tandem

system has two modules in series, the boost pressure performance for both single and tandem module UHS HGVFP systems are consistent with pump literature for liquid-only flows.

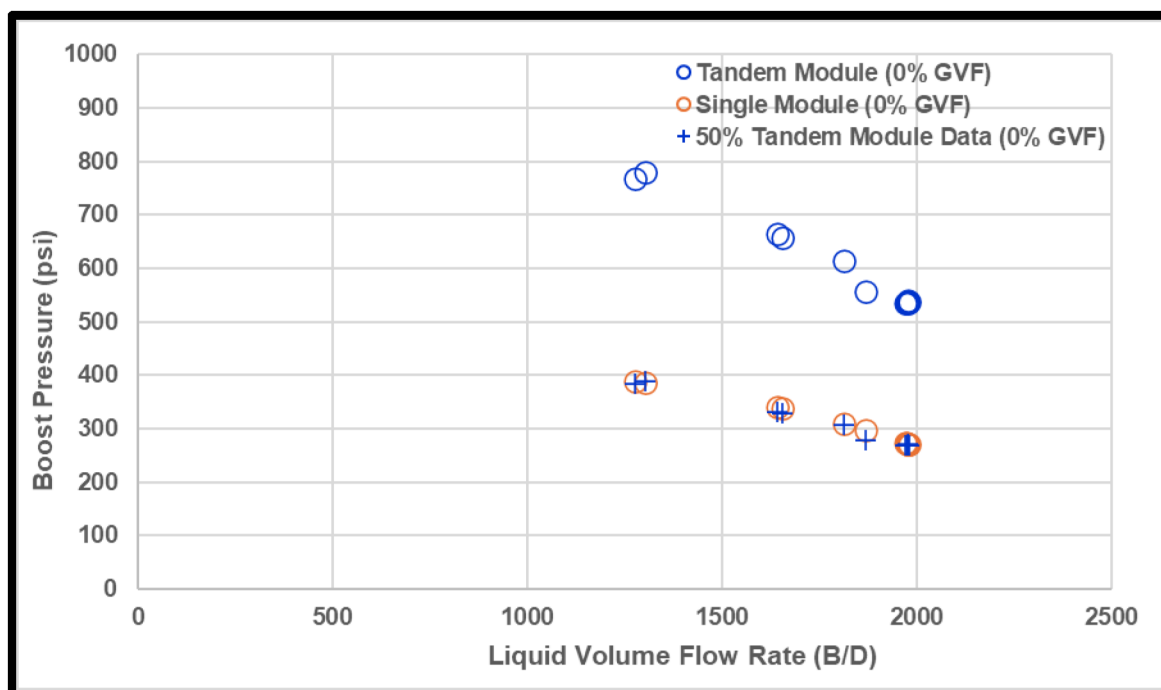


Figure 4—Comparing tandem and single module UHS HGVFP performance at 10,000 rev/min (GVF = 0%; intake pressure = 35psi).

Tandem and Single Module Performance Comparison at Intake GVF > 0%

Figures 5, 6 and 7 compare the boost pressure capabilities of the tandem and single modules at intake GVFs of 15%, 45% and 90%, respectively. The results for other intake GVFs greater than zero (shown in Figure 3) are not presented since the trends are similar although with varying magnitudes of boost pressure at the different liquid volume flow rates.

Figure 5 shows the boost pressure for the tandem and single module at 15% intake GVF for liquid flow rates between 1220 and 1900 B/D. At 1220 B/D, the tandem and single modules provide about 730 and 400 psi of boost pressure, respectively. The corresponding values at 1900 B/D are, respectively 550 psi and 290 psi. For this flow range the tandem module provides about 1.83 to 1.88 times the boost pressure of the single module. The results indicate that unlike the plot in Figure 3, at 15% intake GVF, doubling the number of modules does not double the boost pressure.

Figure 6 shows the comparison for both systems at 45% intake GVF and liquid volume flow rate between 1080 and 1700 B/D. The Figure also shows that the boost pressure boost for each system decreases slower with liquid volume flow rate compared with the test data for the 15% intake GVF. A similar analysis as in Figure 5 shows that the tandem module pressure boost is about 1.68 to 1.75 times those of the single module system. Again, similar to the 15% GVF in Figure 5, doubling the number of modules does not double the boost pressure capability of the UHS HGVFP. In fact, the values observed in Figure 6 are lower than those observed at 15% intake GVF. This indicates increased significant effect of the gas on the pressure boosting capability of the pump at higher intake GVFs.

Figure 7 shows the tandem and single module performance at 90% intake GVF for liquid volume flow rates between 550 and 890 B/D. The single module shows a relatively flat variation of boost pressure with liquid volume flow rate. The tandem module on the other hand still has a negative slope but much lower than the 15% and 45% intake GVFs discussed previously. The boost pressure from the tandem module is about 1.24 to 1.72 times the magnitudes from the single module for this range of liquid volume flow rates.

The values of the boost pressure ratios at 90% intake GVF also suggests that doubling the number of UHS HGVFP modules does not double the boost pressure.

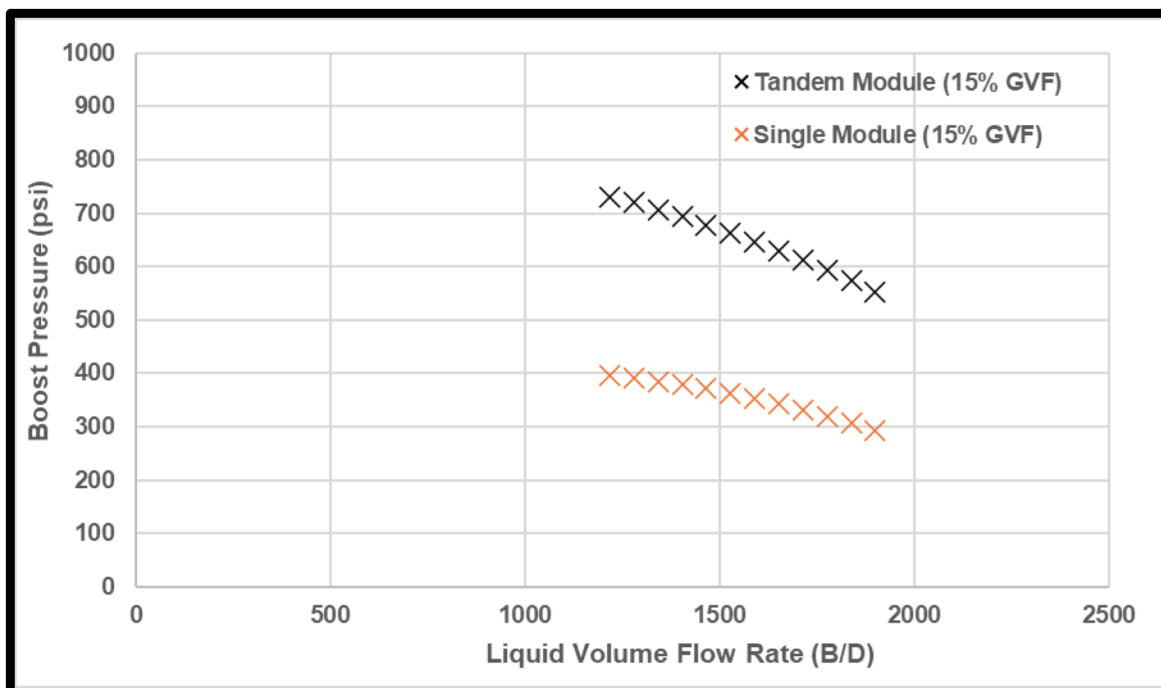


Figure 5—Comparing tandem and single module UHS HGVFP performance at 10,000 rev/min (GVF = 15%; intake pressure = 35psi).

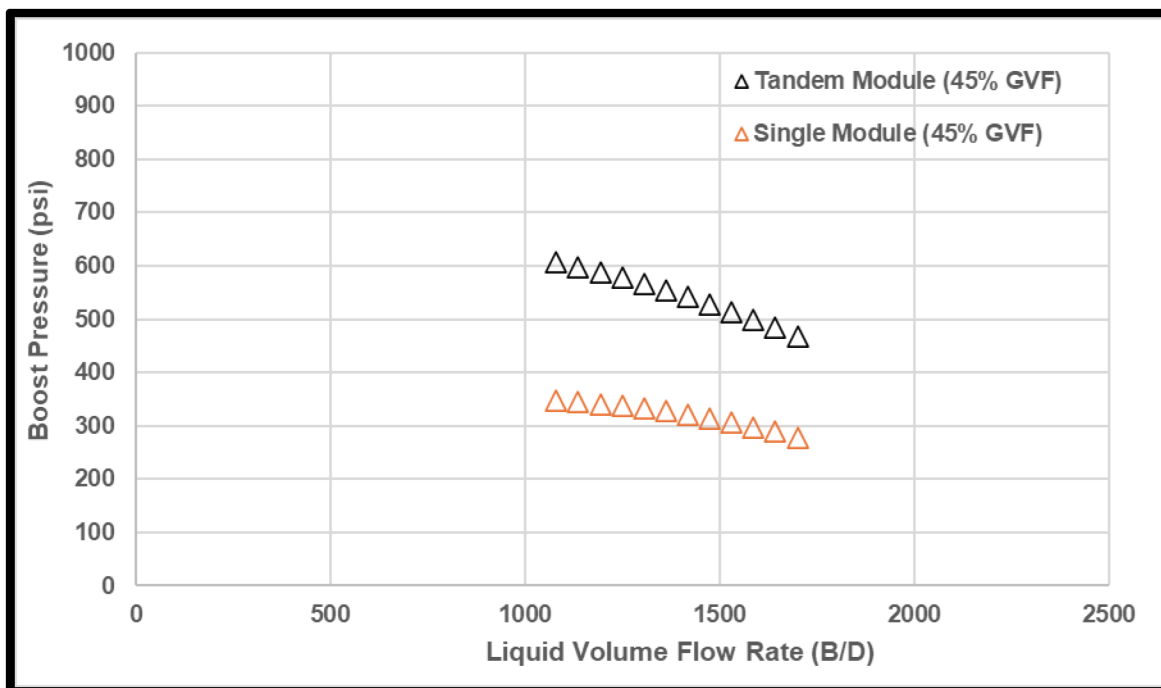


Figure 6—Comparing tandem and single module UHS HGVFP performance at 10,000 rev/min (GVF = 45%; intake pressure = 35psi).

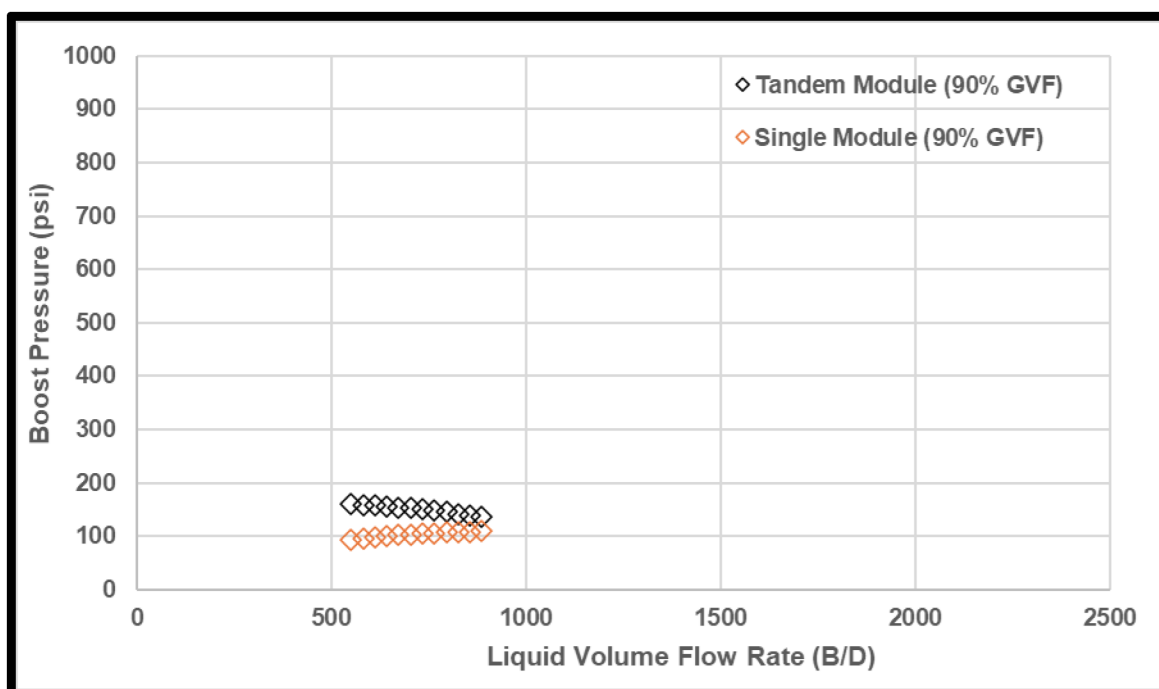


Figure 7—Comparing tandem and single module UHS HGVFP performance at 10,000 rev/min (GVF = 90%; intake pressure = 35psi).

From Figures 5, 6, and 7, it has been established that the effect of the gas when pumping multiphase flows is such that the boost pressure from the additional module is not additive, therefore all the expected boost pressure is not manifested at the pump discharge. This may be explained based on an energy point of view from Equation (1). In the absence of gas all the fluid hydraulic energy is transferred to the liquid and raises the pressure of the liquid to attain the required pump boost pressure. However, when gas is present in the pump, some of the energy is consumed in compressing the gas stream, and the remainder is used to increase the pressure of the liquid stream. This distribution of the useful power is reflected as a reduction in the full pressure boosting of the liquid stream by the pump.

The results presented in this write-up for the tandem UHS HGVFP are a necessary step in understanding the pump behavior for effective design and sizing. Given the non-additive trend due to the presence of gas in the mixture, it is understood that more modules will be added to the pumping system to meet desired boost pressure requirement for a given target liquid volume flow rate. Overall, the gas tolerance capability of the single module UHS HGVFP is high having provided significant positive boost pressure even at 90% intake GVF. This strong attribute of the pump architecture is beneficial since only a relatively small number of additional modules will be needed to meet any field boost pressure requirement for operation in very challenging, gassy or high GOR well conditions.

Summary and Conclusions

The current study highlights the liquid-only and multiphase boost pressure capability of a tandem (two-module) UHS HGVFP and compares it with those of a previously tested single-module UHS HGVFP. The housing outer diameters of both pumps was 4 in, with operating rotational speed of 10,000 rev/min. Air and water were used as the test fluids, with pump intake GVFs varying from 0 to 90%. The average pump intake pressures and temperature during the tests were about 35 psi and about 74°F, respectively.

From the results, when pumping water, the tandem module developed twice the boost pressure of the single module. One-half of the tandem pump boost pressure data overlapped completely on the single module pump data. This indicated that the results of both the tandem and single module pumps were

consistent with pump-in-series behavior from literature. When handling multiphase flows, the tandem pump boost pressure decreased with increasing intake GVF, but still provided significant boost pressure at 90% intake GVF. Comparing the tandem pump with the single module at different intake GVFs shows that the boost pressure ratio by the tandem pump varied between 1.83 to 1.88 at 15% intake GVF between 1220-1900 B/D, and about 1.24 to 1.72 at 90% intake GVF between 550-890 B/D. The overall trend is that the boost pressure is not additive when pumping gas-liquid mixtures since some of the hydraulic energy is expended to compress the gas stream. Nevertheless, the tandem UHS HGVFP still demonstrated a high resilience and tolerance to gas flow with intake GVFs of 90%.

The current work is beneficial to appropriately design and size a multiphase pump especially for field installations, where pumps will be operating in very gassy or high GOR environments. The results form part of a database for determining the effective number of modules required to satisfy a given boost pressure and flow rate requirement in the field. The end objective is to ensure installation of an UHS HGVFP with sufficient lifting capacity to significantly increase the full production potential of any gassy or high GOR well.

Conversion Factor Table

1 ft	0.3048 m
1 in.	0.0254 m
1 B/D	0.117348 m ³ /d
1 lbm	0.453592 kg
1 lbm/ft ³	16.0185 kg/m ³
1 psi	6894.76 Pa
t°F	[{(t°F)-32}/1.8]°C
1 hp	745.7 W

Nomenclature

- n Polytropic index (dimensionless)
- P₁ Absolute pressure of fluid mixture at pump intake, MLT^{-2} , psi [Pa]
- P₂ Absolute pressure of fluid mixture at pump discharge, MLT^{-2} , psi [Pa]
- P_{mix} Useful power transferred to a fluid mixture, ML^2T^{-3} , hp [W]
- Q_L Volume flow rate of the liquid, L^3T^{-1} , B/D or ft³/s [m³/s]
- Q_{G1} Volume flow rate of the gas at pump intake conditions, L^3T^{-1} , B/D or ft³/s [m³/s]

ESP Electric Submersible Pump

GOR Gas Oil Ratio

GVF Gas Volume Fraction

RTD Resistance Temperature Detector

TTESP Through-Tubing Electric Submersible Pump

UHS HGVFP Ultra-High-Speed High Gas Volume Fraction Pump

VSD Variable Speed Drive

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